

Project Workflow: Week 1

Phase: Project Planning & Concept Finalization

1. Objective

The primary goal of Week 1 was to establish a robust project foundation for recreating the classic Bounce Ball game with simple obstacles and levels. This included:

- Defining the project scope and core gameplay mechanics.
- Selecting an optimal technology stack for GUI rendering and event-driven game logic.
- Conceptualizing the level structure and visual design to align with classic arcade standards.

2. Detailed Task Breakdown

Feature Identification

Conducted research on classic Bounce Ball / Brick Breaker style games to finalize the core feature set: ball movement physics, paddle/obstacle collision detection, multiple levels, a score system, and a game-over screen.

Technical Stack Selection

Selected Java with Object-Oriented Programming principles for clean state management, Swing for GUI rendering, the Timer class for the game loop, and Exception Handling for robust runtime stability.

UI/UX Design

Created wireframes for the game window layout, including the play area, obstacle placement, score display, and game-over screen, establishing a simple and responsive arcade-style visual identity.

Architectural Setup

Designed a modular, object-oriented class architecture separating the game loop, ball physics, obstacle/collision logic, and level management to ensure clean code separation between rendering and game logic.

3. Deliverables

- Project Scope Document: A formal outline of the game's features and limitations.
- UI Wireframe Sketches: Conceptual layouts for the game window and HUD elements.
- Level Design Outline: Initial plan for obstacle placement across multiple levels.
- Project Timeline: A structured 10-week development roadmap.

4. Tools & Resources

Development Environment

- Java JDK 11+
- IntelliJ IDEA / VS Code

Reference Materials

Java Swing, AWT, and Timer-based game loop references.